MIL-E-1/1180C
17 June 1970
SUPERSEDING
MIL-E-1/1180B
3 November 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6781

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, oscillator integral cavity, frequency range 8,500 to 10,000 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	F	T(tuner plate)	Alt
Unit:	V	Vdc	Vdc	mAdc	MHz	°C	ft
Maximum:	6.9	350	-1,000	42	10,000	200	60,000
Minimum:	5.7	-20	---	---	8,500	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Weight: 8 ounces

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	F	T (tuner plate)
Unit:	V	Vdc	Vdc	MHz	°C
	6.3	200	Adjust for maximum Po	9,300 ± 0.3%	(See notes 2, 3, and 11)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1					
1231	Emission	10	Ef = 5.7 V	ΔIk/Ik	---	-15	%
1256	Electrode current (cathode)	---	F = 9,660 MHz ±0.3%; Er (mode 6-3/4) max Po	Ik	---	23.0	mAdc
4213	Reflector voltage (1)	---	F = 9,000 MHz ± 0.3%	Er	-75	-135	Vdc
4213	Reflector voltage (2)	---	F = 9,660 MHz ± 0.3%	Er	-100	-165	Vdc
4229	Total reflector current	10	t = 120	Ir	---	3.0	μAdc
4223	Mechanical tuning range	---		F	8,500	10,000	MHz
4223	Tuning rate	---	F = 8,500 to 10,000 MHz	Turns	1-1/4	1-3/4	Turns
4250	Power output (1)	---	F = 9,000 MHz ± 0.3%; Er (mode 6-3/4) -125 V; VSWR, not less than 1.5 in any phase position	Po	15	---	mW
4250	Power output (2)	---	F = 9,660 MHz ± 0.3%; Er (mode 6-3/4) -125 V; VSWR, not less than 1.5 in any phase position	Po	15	---	mW
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	4, 6	F = 60 ± 2 Hz; 10 G; t = 120	Ik	0	10	μAdc
1211	Insulation of electrodes	---	300 Vdc, tube cold	Rh-rs Rk-rs	2 2	--- ---	Meg Meg
1301	Heater current	---		If	1.08	1.32	A
4250	Power output (3)	12	F = 8,500 MHz; F = 9,300 MHz; F = 10,000 MHz; Er (mode 5-3/4) max Po; Ers = 300 Vdc	Po	40	---	mW
4280	Electronic tuning range	5	F = 8,940 to 9,660 MHz; Er (mode 6-3/4) 50% max Po	ΔF	20	---	MHz
4231	Hysteresis	---	Er (mode 5-3/4) max Po; F = 9,300 MHz ± 0.3%	Ratio	---	0.5	---
1105	Permanence of marking	---		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
4551	Life tests	---	Group C: power output (1)	t	500	---	hrs
---	Life-test end points	---					
4250	Power output (1)	---	F = 9,000 MHz ± 0.3%; Er (mode 6-3/4) -125 V, VSWR, not less than 1.5 in any phase position	Po Ir	12 ---	--- 10	mW µAdc
4229	Total reflector current	---	t = 120 sec	Ir	---	10	µAdc
	Periodic-check tests						
1031	Sweep-frequency vibration	4, 6, 7		ΔF	---	4.0	MHz
1041	Shock	8	100 G; duration 4 ms; power output (1)	ΔF	---	4.0	MHz
---	Heater voltage coefficient	---	Ef = 5.7 to 7.0 Vac	ΔF/ΔEf	---	4.0	MHz/V
4027	Temperature coefficient	---	Er (mode 6-3/4); TA = -35° to +90°C	ΔF/°C	---	-0.20	MHz/°C
---	Low pressure	9	Er (mode 6-3/4)	ΔF	---	2.0	MHz
---	Frequency modulation	---	F = 9,660 MHz ± 0.3%; Er (mode 6-3/4); Ef = 5.7 to 7.0 Vdc	ΔF	---	0.1	MHz

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.
2. All oscillation tests shall be made with the tube rigidly connected to a UG-39/U or a UG-40A/U flange on appropriate RG-52/U waveguide equipment. The load VSWR for the tube shall be less than 1.1:1.
3. Forced-air cooling is required above 10 watts resonator power input.
4. During vibration, the reflector current shall be monitored with a recorder having a frequency response from dc to at least 100 Hz and sufficient sensitivity to give a reasonable indication of current within the specified limit. There shall be no reflector current bursts greater than the specified amount.
5. The power output shall have no discontinuities between half-power points for either direction of reflector voltage change.
6. Vibrate once along the 3.000-inch maximum dimension and once along the 1.750-inch maximum dimension.

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NOTES: -Continued

7. The change in frequency (ΔF) during vibration shall be measured peak-to-peak.
8. The tube shall be given five shocks in each of three mutually perpendicular planes, one of which shall be along the 3.000-inch maximum dimension.
9. The frequency shall be stabilized at a pressure of 55 mmHg. The pressure shall be increased to 750 mmHg and the frequency at this pressure read within the specified time. The resulting frequency change shall not exceed the specified limit.
10. The tube shall not be oscillating during this test.
11. Temperature to be measured at tuner plate.
12. The tube shall meet the minimum power output requirement when operating into a matched load.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Air Force - 11, 80
DSA - ES

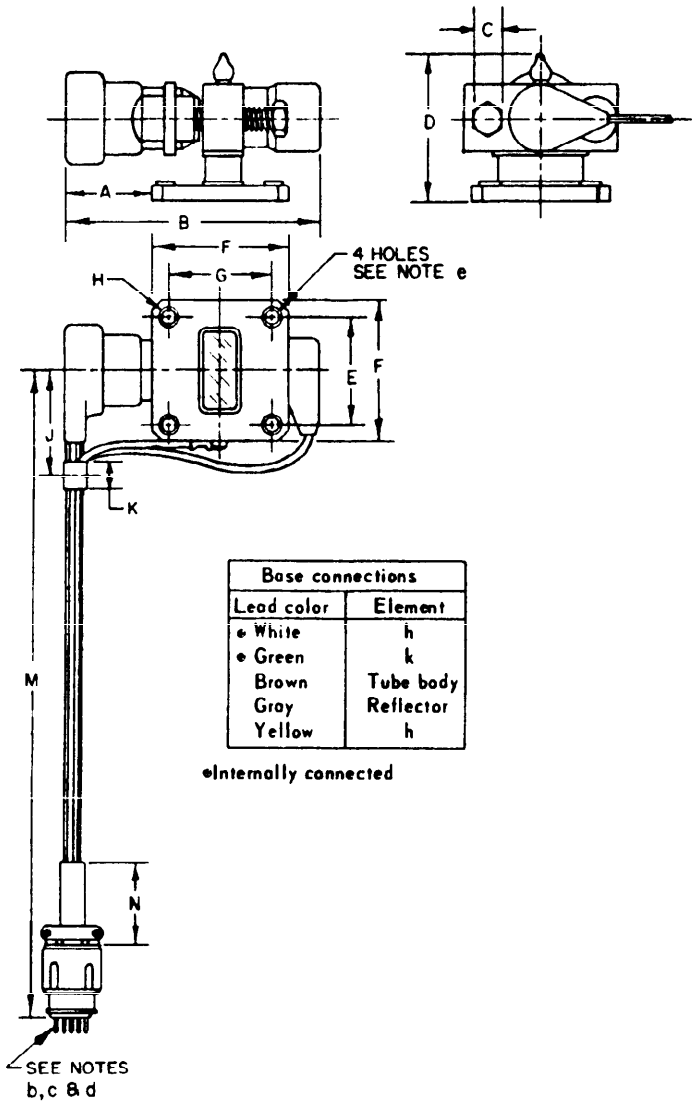
User activities:
Army - ME, WC
Navy - AS, OS, SH, CG

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2409)

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Base connections	
Lead color	Element
• White	h
• Green	k
Brown	Tube body
Gray	Reflector
Yellow	h

•Internally connected

FIGURE 1. Outline drawing of electron tube type 6781.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note a)		
A		1.000 (25.40)
B		3.000 (76.20)
M	6.750 (171.45)	7.250 (184.15)
Quality conformance inspection, part 2		
D		1.750 (44.45)
E	1.216 (30.89)	1.224 (31.09)
G	1.276 (32.41)	1.284 (32.61)
H	.214 (5.44)	.224 (5.69)
Quality conformance inspection, part 3 periodic-check		
C	.365 (9.27)	.377 (9.58)
F	1.640 (41.66)	
J	1.500 (38.10)	
K	.500 (12.70)	
N	3.000 (76.20)	

NOTES:
a. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
b. Plug connections are as follows:
   A - White D - Gray
   B - Yellow E - Green
   C - Brown
c. Moisture - fungus proof varnish, MIL-V-173, shall be applied over soldered joints in connector.
d. Connector mates with Viking, or equivalent, receptacle VP5/2AA1 including hardware for panel mounting, or with receptacle VP5/2AA1 and VS7/23C1, less hardware, for cable connection.
e. With removable inserts .185 ±.005 (4.70 ±.13 mm) diameter.
f. Mounting: Mates with UG-39/U or UG-40A/U flange. Mounts in any position.
g. Tuner: Single screw.

FIGURE 1. Outline drawing of electron tube type 6781 - Continued.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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